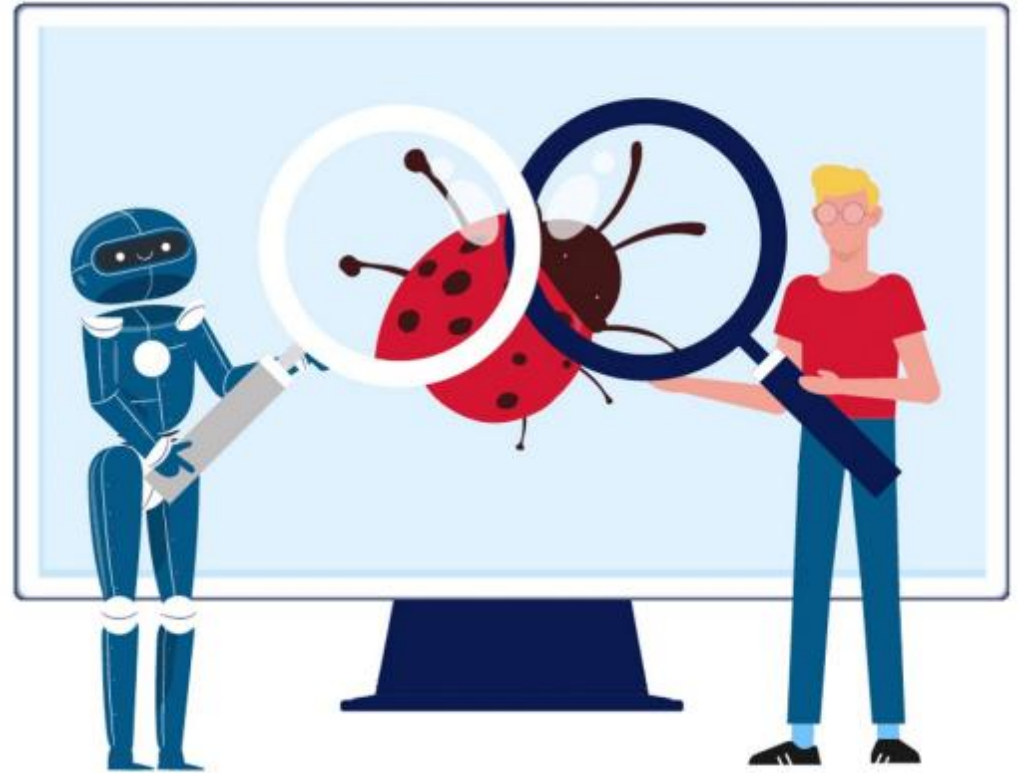


Automated Software Testing

Spring, 2026

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1. Why we need Automated Testing

- 1) Manual software testing is **slow, error-prone, and hard to repeat accurately**:
 - Testing All workflows/all fields/all negative scenarios is time consuming
 - Become boring and hence error prone as human testers lose concentration.
 - The testing finishes for the day when the testers go home

- 2) Yet software testing needs to be **fast, accurate, and repeatable**:
 - Performed frequently without a time penalty.
 - Test results can be relied on as a quality indicator.
 - Be repeatable to allow for regression testing.

2. What is Automated Testing?

It's a software testing method that compares expected and actual results of test cases with the help of special automation testing tools

- Automated execution of tests, or collections of tests
- Automated collection of results
- Automated evaluation of results
- Automated reports
- Automated measurement of test coverage

The Key Stages of Automated Testing

Consider the following steps:

1. Define your goals and create your strategy in automation testing;
2. Choose automation testing tools to execute a test task like Selenium, Egg Plant, etc.;
3. Set up test environment;
4. Develop test scripts on the basis of testing requirements;
5. **Test execution and result analysis.**

The Key Differences Between Manual and Automated Testing

Benefits and drawbacks automated testing has :

Pros: (High ROI & Faster GoTo market)

- Supports execution of **repeated** Test Cases
- Aids in testing a **large** Test Matrix
- Enables **parallel** execution
- Encourages **unattended** execution
- Improves **accuracy** thereby reducing human-generated errors
- Saves **time** and money

Cons:

- Automation instruments are usually **expensive**;
- It is ineffective in testing **user experience** in applications;
- **Coding** knowledge and experience are a must.

The Key Differences Between Manual and Automated Testing

Manual testing is commonly used when:

- You have a **short-term project** with a low budget;
- You need to complete **exploratory testing**;
- You are going to run **ad-hoc testing**, which is usually unplanned, and gathering testing insights are important in this process;
- You should test app **usability** and measure how valuable the user experience is for your end users.

The Key Differences Between Manual and Automated Testing

Automation testing is suitable when:

- You know a specific number of **regression tests** for your project;
- You should test server models and web servers in **load and stress testing**;
- You have a **big project** where it's essential to test several software functionalities;
- You run **performance testing** to check software quality attributes like scalability, reliability, speed, etc.

3. Selenium

- Selenium is one of the most popular **Automated Testing suites**.
- Selenium is designed to support and encourage Automation Testing of **functional** aspects of **web-based applications** and a wide range of browsers and platforms.

- It's an open-source
- It has a large user base and helping communities
- It has multi-browser and platform compatibility
- It has active repository developments
- It supports multiple language implementations

Selenium

Selenium automates browsers.

<https://www.selenium.dev>

Primarily it is for automating web applications for testing purposes, but is certainly not limited to just that.

Getting Started



Selenium WebDriver

If you want to create robust, browser-based regression automation suites and tests, scale and distribute scripts across many environments, then you want to use Selenium WebDriver, a collection of language specific bindings to drive a browser - the way it is meant to be driven.



Selenium IDE

If you want to create quick bug reproduction scripts, create scripts to aid in automation-aided exploratory testing, then you want to use Selenium IDE; a Chrome, Firefox and Edge add-on that will do simple record-and-playback of interactions with the browser.

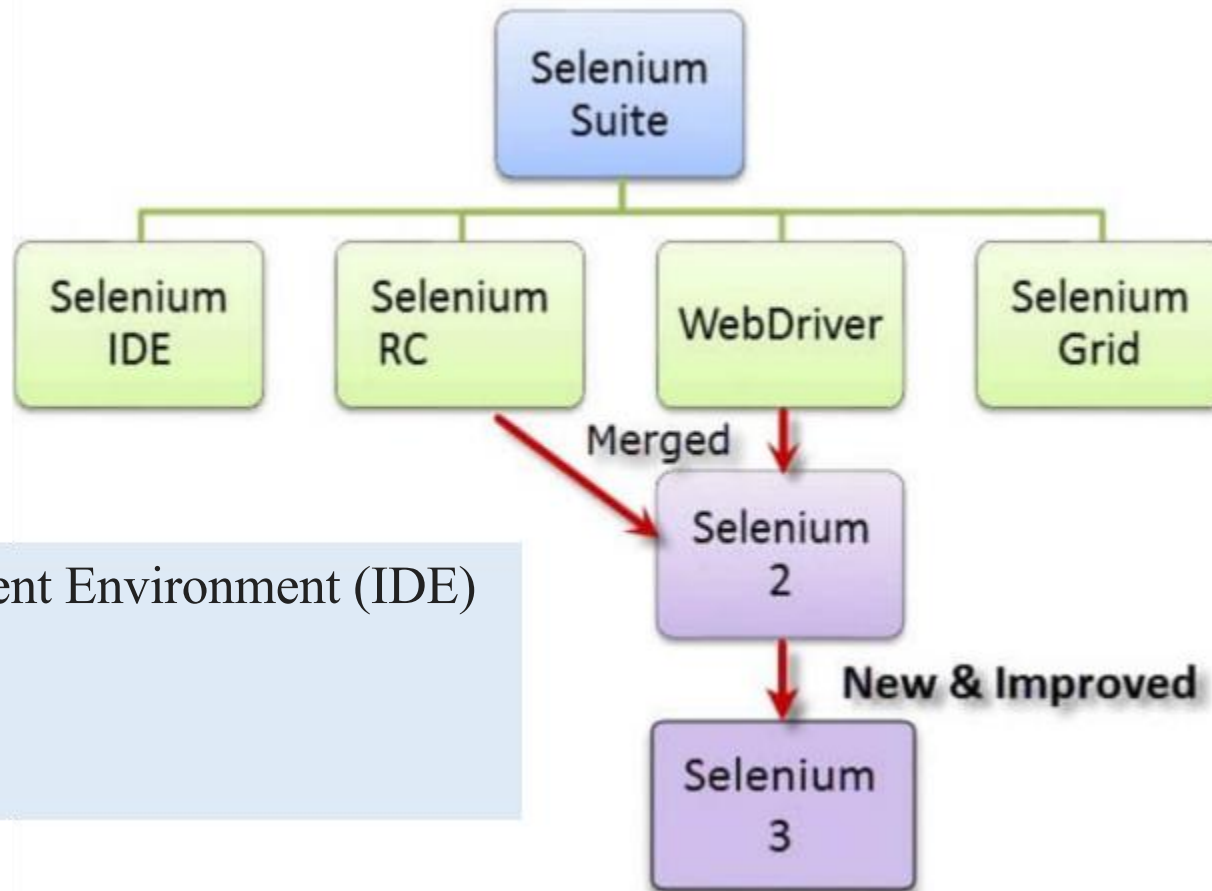


Selenium Grid

If you want to scale by distributing and running tests on several machines and manage multiple environments from a central point, making it easy to run the tests against a vast combination of browsers/OS, then you want to use Selenium Grid.

Selenium Components

- Selenium is a package of several **testing tools**, hence it is referred to as a Suite.
- Each of these tools is designed to cater to different testing and **test_environment requirements**.



- Selenium Integrated Development Environment (IDE)
- Selenium Remote Control (RC)
- WebDriver
- Selenium Grid

Selenium Components --- Selenium Core

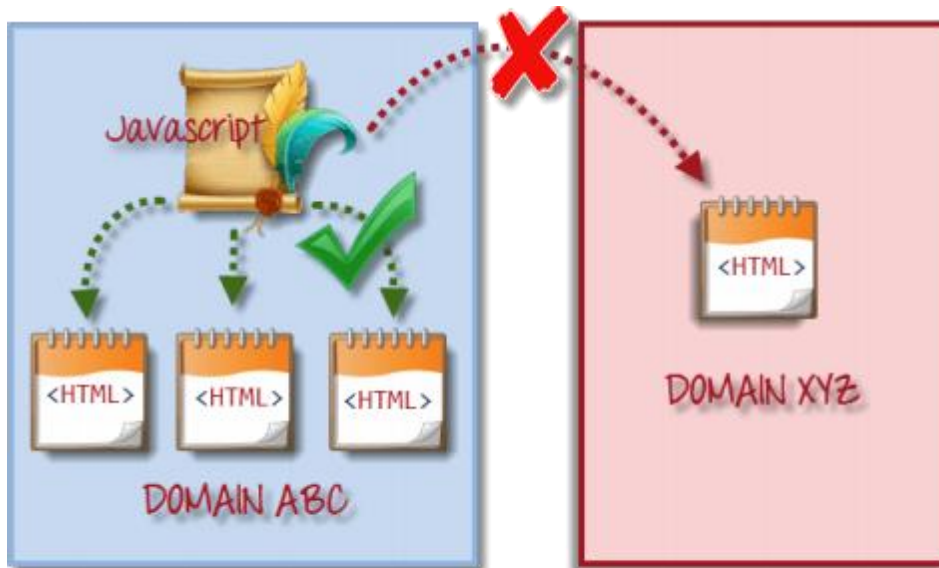
- Selenium was created by Jason Huggins in 2004 at ThoughtWorks.
- As repetitious Manual Testing of their application was becoming more and more inefficient, a JavaScript program that would automatically control the browser's actions was created.
- First named as the “**JavaScriptTestRunner.**”
- Open-source and later re-named as **Selenium Core.**

Selenium IDE (Selenium Integrated Development Environment)

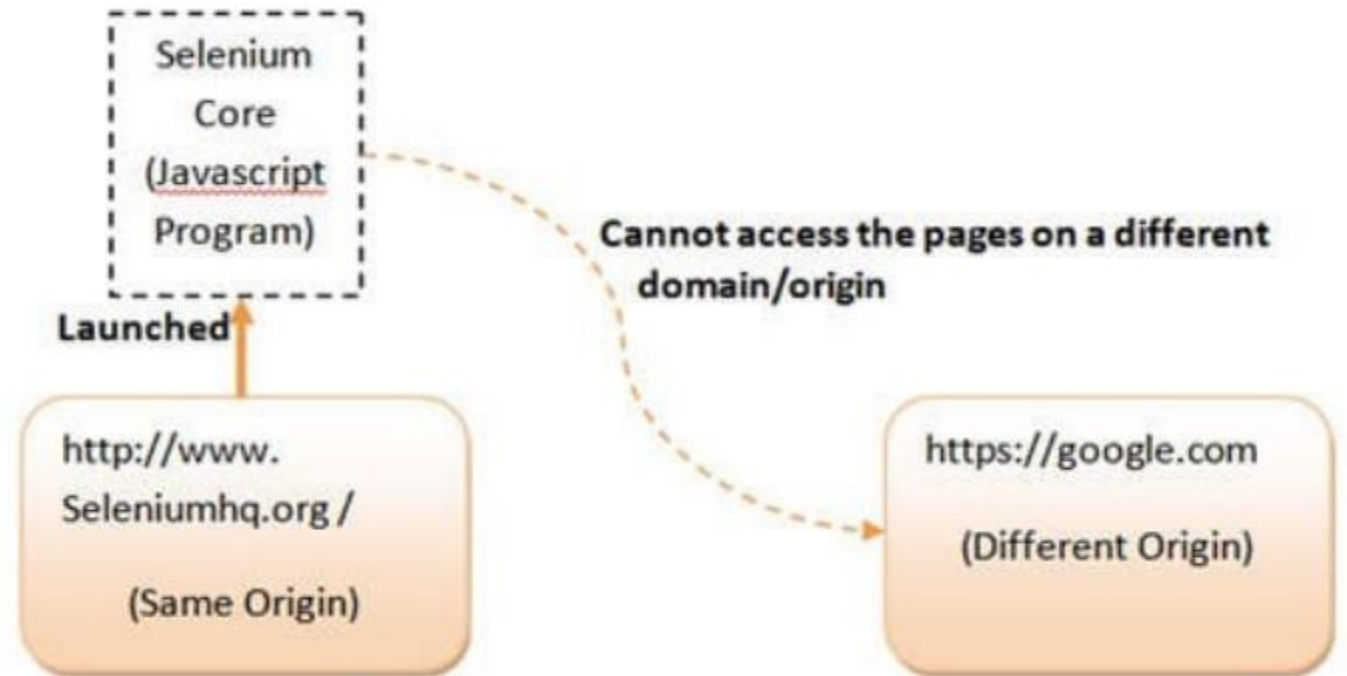
- Shinya Kasatani of Japan created Selenium IDE, a Firefox extension that can automate the browser through a **record-and-playback** feature.
 - Further increase the speed in creating test cases.
 - The Simplest framework in the Selenium suite and the easiest one to learn.
- Doesn't support **iterations** and **conditional** statements
 - Doesn't support **loops**
 - Doesn't support **error handling**
 - Doesn't support test **script dependency**

Selenium Remote Control (Selenium RC)

- Testers using Selenium Core had to install the whole application under test and the web server on their own local computers because of the restrictions imposed by the **same origin policy**.

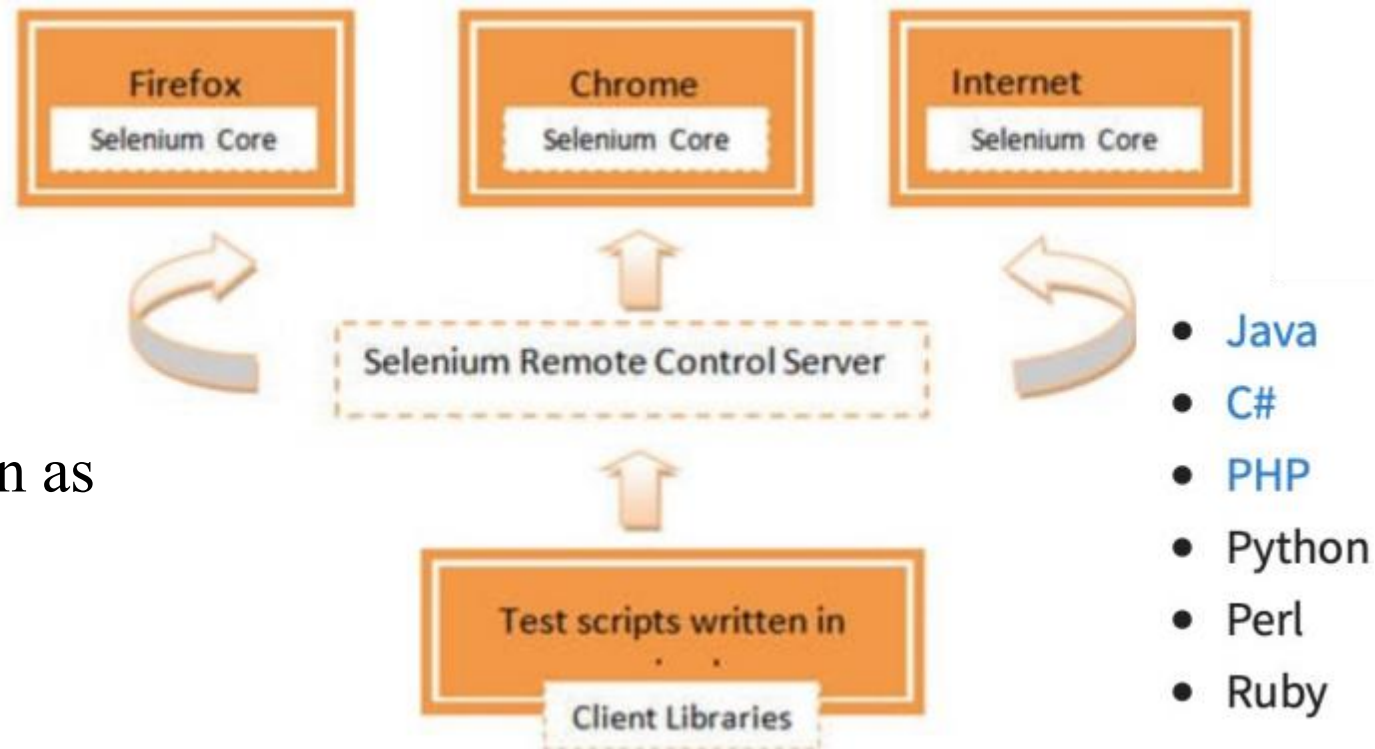


under same origin policy, a Javascript program can only access pages on the same domain where it belongs. it cannot access pages from different domains



Selenium Remote Control (Selenium RC)

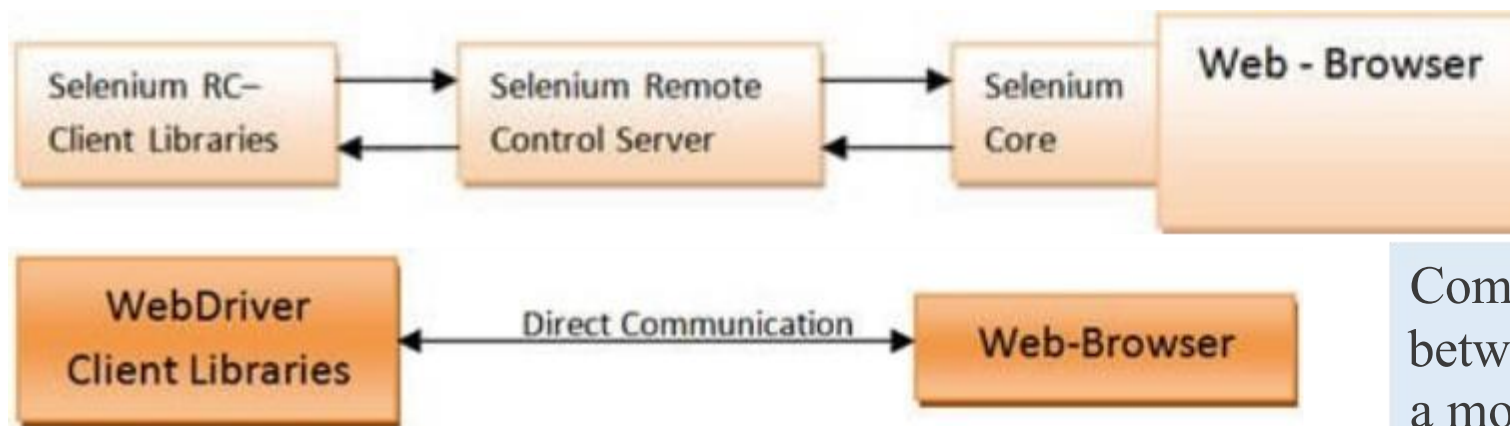
- ThoughtWork's engineer, Paul Hammant, created a tool written in Java to allow a user to construct test scripts for a web-based application in **any programming language he/she chooses**.
- Selenium RC came as a result to overcome the various disadvantages incurred by *Selenium IDE or Core*.



This system became known as the Selenium Remote Control or **Selenium 1**.

Selenium WebDriver

- WebDriver allows a user to perform web-based automation testing. WebDriver is a different tool altogether that has various advantages over Selenium RC.
 - **Directly communicates with the web browser** from the OS level and uses its native compatibility to automate, which is **faster** than other tools of Selenium
 - Supports a **wide range of web browsers**, programming languages and test environments.
 - Supports efficient handling mechanisms for **complex user actions** like dealing with dropdowns, Ajax calls, switching between windows, navigation, handling alerts etc.



Compatibility analysis between **Selenium RC** and **WebDriver**, a more powerful **Selenium 2**

Selenium Grid

- Selenium Grid is a tool to run **parallel tests** across different machines and different browsers all at the same time. Parallel execution means running multiple tests at once.
- Features:
 - Enables simultaneous running of tests in multiple browsers and environments.
 - Saves time enormously.
 - Utilizes the hub-and-nodes concept. The hub acts as a central source of Selenium commands to each node connected to it.

How to Choose the Right Selenium Tool

Selenium IDE

- To learn about concepts on automated testing and Selenium, including:
 - Selenese commands such as type, open, clickAndWait, assert, verify, etc.
 - Locators such as id, name, xpath, css selector, etc.
 - Executing customized JavaScript code using runScript
 - Exporting test cases in various formats.
- To create tests with little or no prior knowledge in programming.
- To create simple test cases and test suites that you can export later to RC or WebDriver.
- To test a web application against Firefox and Chrome only.

How to Choose the Right Selenium Tool

WebDriver

- To use a certain programming language in designing your test case.
- To test applications that are rich in AJAX-based functionalities.
- To execute tests on the HtmlUnit browser.
- To create customized test results.

Selenium Grid

- To run your Selenium RC scripts in multiple browsers and operating systems simultaneously.
- To run a huge test suite, that needs to complete in the soonest time possible.

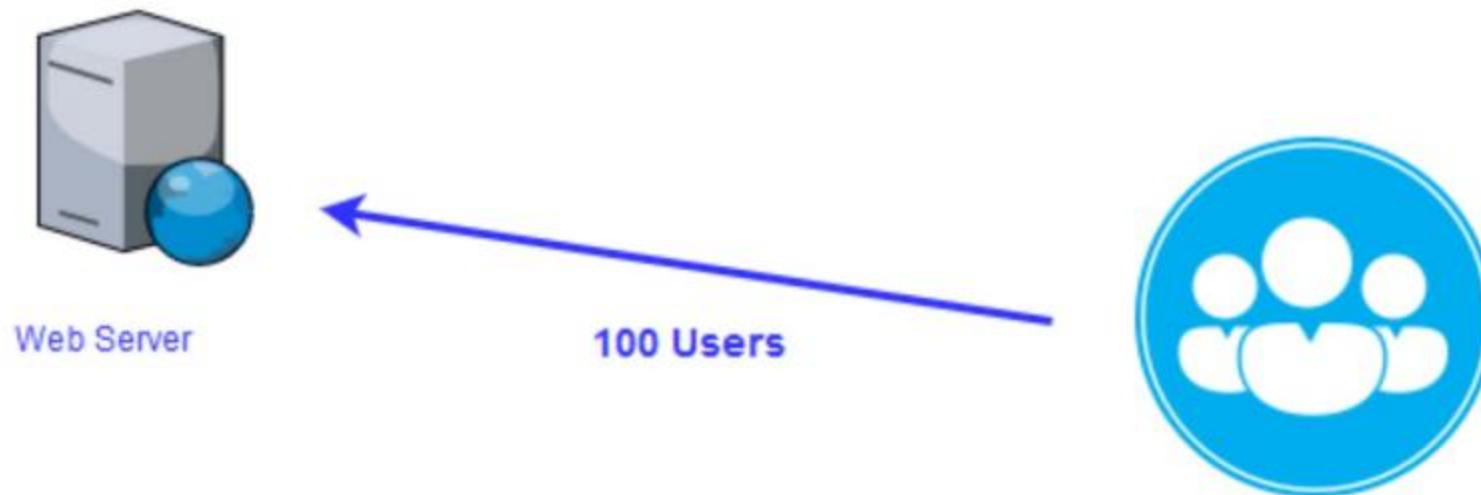
4. Jmeter

- ‘**Apache JMeter**’ is an **open source**, 100% **java-based application** with a graphical interface, which can analyze and measure the performance of web application or a variety of services.

<https://jmeter.apache.org>



- Developed by Stefano Mazzocchi of the Apache Software Foundation.
 - Primarily written to test the performance of Apache JServ(currently known as Apache Tomcat project).



JMeter Features

- **Open source application**
- **User-friendly GUI :** simple and interactive GUI.
- **Support various testing approach:** like *Load Testing, Distributed Testing, and Functional Testing*, etc.
- **Platform independent:** run on any environment /workstation that accepts a *Java virtual machine*
- **Support various server types:** highly extensible for different server : *Web, Database, Mail...*
- **Support multi-protocol:** protocols such as *HTTP, JDBC, LDAP, SOAP, JMS, and FTP...*
- **Simulation:** using *virtual users* to generate heavy load against web application under test.
- **Framework:** *multi-threading* framework which allows concurrent and simultaneous sampling of different functions by many or separate thread groups.
- **Remote distributed testing:** *Master-Slave* concept for distributed testing where master will distribute tests among all slaves and slaves will execute scripts against your server.
- **Test result visualization:** viewed in different formats like *graph, table, tree, and report etc.*